KVL CRM - Gap Analysis

Message (WhatsApp/SMS), Call aur Email system: gaps aur unka BUILD STATUS (updated)

Ye list code check karke bani hai. "Nahi hai" wale points grep se confirm kiye gaye hain; kuch cheezein kisi aur jagah ho sakti hain jo check mein nahi mili. Is se pehle 90-Day Follow-up Engine aur own-number WhatsApp (Cloud API / QR) already ban chuke hain (commit 98e68a5). **Neeche section 0 mein dikhaya gaya hai ki PDF ke har gap ke liye ab kya bana diya gaya hai.**

0. Build status: har gap ke liye kya bana

Sab code ban chuka hai: TypeScript clean, 256 unit tests pass, production build pass. **Abhi live nahi hai:** database schema chalani hai, keys/webhooks set karne hain, aur asli WhatsApp/Twilio/Resend se live test baaki hai (docs/COMMS_SETUP.md).

Gap (PDF)	Ab kya bana hai	Kahan
1. Central DNC list	Har WhatsApp / SMS / email / call bhejne se pehle check. STOP, unsubscribe, hard bounce, spam complaint, do-not-call outcome khud list mein. Manual add/remove.	lib/compliance, Comms Hub > Settings
2. Email reply	Resend webhook (email.received) + generic /api/email/inbound. Reply aate hi drip ruke, team alert, auto-reply/out-of-office ignore.	lib/messaging/emailEvents.ts
3. Delivery status	Twilio StatusCallback aur Meta statuses: sent, delivered, read, failed (kabhi downgrade nahi). Mock alag dikhta hai. Fail par team alert.	lib/messaging/status.ts
4. Reply par instant alert	Notification bell (real, 30s poll) + optional WhatsApp/email alert team ko. Payment-intent, 24h overdue, failures par high priority.	lib/notify, TopNav
5. Email deliverability	Unsubscribe link + one-click headers, bounce/complaint handling, SPF/DKIM/DMARC guide.	docs/COMMS_SETUP.md
Message: template manager	WhatsApp (Meta) templates, email templates aur quick replies ek jagah; approved template se cold message.	Comms Hub > Templates
Message: media	Image / PDF / video bhejna (Cloud API, QR, Twilio).	send.ts, Inbox
Message: AI auto-answer, business hours, assignment	Knowledge base se AI jawab (kabhi facts invent nahi), off-hours auto-reply, 6h cooldown, chat auto-assign (owner ya sabse kam load).	lib/comms
Call: dialer, outcome, callback	Click-to-call (pehle aapka phone, phir lead), 8 outcomes, callback reminder, not-picked par WhatsApp fallback, 3 miss par task.	lib/telephony, Comms Hub > Calls
Call: recording, voicemail	Recording playback + consent notice. Voicemail drop sequence mein.	Calls tab, followup sequence
Email: editor, tracking, inbox	Email templates, open/click counts, thread inbox with reply.	Comms Hub > Email
Follow-up: editor, links, reports, timezone	Sequence editor (nayi leads ke liye; chalti leads ka snapshot safe), {{bookingLink}} / {{paymentLink}}, ready-to-pay par payment link auto-send, reports, lead ka local-time quiet hours.	Sequence + Reports tabs
Reminder Engine	Task due + 24h overdue escalation, meeting 24h/2h/15m client reminder, invoice pre-due/due/+3/+7/+15, callback ('kal 5 baje') se auto reminder. DNC + quiet hours + done-check.	lib/reminders, Reminders tab

Jo abhi nahi hai: Meta WhatsApp template approval Meta se karwana padega; Resend inbound API ka exact format docs se verify karna hai (fallback endpoint ready hai); no-show par reschedule message auto nahi jata

(team ko prompt jata hai, galat message se bachne ke liye).

1. Sabse zaroori gaps (inke bina 100% nahi)

#	Gap	Kyon problem hai
1	Central opt-out (DNC) list nahi hai. Client STOP likhe to sirf follow-up band hota hai; birthday, webinar reminder aur bulk campaign phir bhi message bhej sakte hain.	Legal risk (TRAI / DPDP) aur WhatsApp number block hone ka risk.
2	Email reply system nahi hai. Resend sirf bhejta hai; inbound email ka webhook CRM mein nahi hai.	Client email par reply kare to drip chalta rehta hai aur 24h AI-nudge ka timer shuru nahi hota.
3	Delivery status ignore hota hai. Twilio / Meta ke delivered, read, failed events handle nahi hote.	System "sent" maan leta hai chahe message pahuncha na ho. Bina keys ke mock bhi "sent" count hota hai.
4	Naye reply par team ko turant alert nahi milta. Task sirf "payment ready" ya "24h overdue" par banta hai.	Team ko 24 ghante baad hi pata chalta hai ki client ne reply kiya tha.
5	Email deliverability setup nahi: SPF/DKIM/DMARC guide, bounce aur spam-complaint handling, unsubscribe link.	Email spam mein jayega; bounce hue address par bhejne se sender reputation kharab hogi.

2. Message (WhatsApp / SMS)

- WhatsApp **template manager** nahi hai. Cold lead ke liye approved template zaroori hai; abhi sirf ek env variable hai (WHATSAPP_CLOUD_TEMPLATE_NAME).
- **Image, PDF, brochure** bhejne ka option nahi hai (media_url column hai, UI aur send nahi).
- **Canned replies** aur quick templates nahi hain.
- AI abhi sirf "hum reply karenge" wala holding message bhejta hai. Client ke sawal ka asli jawab (FAQ, pricing) dene wala **AI auto-answer** nahi hai.
- **Business hours auto-reply** aur chat ko team member ko **assign** karna nahi hai.
- India SMS ke liye **DLT** registration abhi manual hai.

3. Call

- App ke andar **click-to-call dialer** nahi hai.
- **Call outcome** log karne ka form nahi hai (connected, not picked, callback, interested). Isse engine ko pata nahi chalta ki call ke baad kya karna hai.
- Call task banta hai, par **callback schedule** karke reminder nahi milta.
- **Call recording playback aur consent** nahi hai. Recording URL save hota hai, dikhta nahi.
- Ringless voicemail ka code hai, par follow-up sequence se juda nahi.
- Do baar call na uthane par **WhatsApp se fallback** nahi hai.

4. Email

- Email section mein sirf campaign list hai (naam, status, date). **Email template editor** nahi hai; templates sirf naam hain.
- **Open aur click tracking** nahi hai, isliye pata nahi chalta kaun interested hai.
- Email **thread inbox** nahi hai (do-tarfa baatcheet ek jagah).

5. Follow-up Engine mein bache hue gaps

- 90 din ki sequence **code mein fixed** hai. UI se edit nahi hoti, aur lead source / stage ke hisaab se alag sequence nahi hai.
- Message mein **booking link** aur **Razorpay payment link** auto nahi judte.
- **Reports** nahi hain: reply rate, kaunsa step convert karta hai, kitne paid hue.
- Lead ka apna **time zone** nahi hai (abhi sab IST).
- WhatsApp cold follow-up ke liye approved template zaroori hai (Meta ka 24-hour rule).

6. Reminder system

Status: pehle reminder system partly tha. Dedicated **Reminder Engine ab ban chuka hai** (section 0 dekhein); neeche table purane status ko dikhata hai.

Reminder	Pehle ka status (ab sab bana hai)
Webinar 24h / 1h pehle (WhatsApp, SMS, email)	Hai aur kaam karta hai (cron har 15 min)
Client follow-up (90 din, WhatsApp, email, call)	Hai (Follow-up Engine)
Team task: call karo, payment link bhejo, 24h reply overdue	Hai (follow-up se task banta hai)
Birthday / seasonal message	Hai (roz cron)
Task due / overdue par team ko notification	Nahi hai. due sirf text field hai
Meeting / demo se pehle client ko reminder	Nahi hai (no-show badhte hain)
Callback reminder (client ne kaha kal 5 baje call karna)	Nahi hai
Invoice overdue reminder	Sirf UI mein dikhta hai, asli mein bhejta nahi
Team members ko push / WhatsApp reminder	Nahi hai (Web Push sirf anonymous visitors ke liye)

Reminder Engine design (ban chuka hai):

- Ek **reminders table** aur ek cron (har 5 minute) jo due reminders bheje.
- **Task / callback reminders:** due time par team member ko WhatsApp, email ya in-app alert; overdue hone par manager ko escalation.
- **Meeting / demo reminders:** client ko 24h, 2h aur 15 min pehle message, aur no-show hone par auto reschedule message.
- **Invoice reminders:** due se 3 din pehle, due date par, aur overdue par 3, 7, 15 din baad.
- **Callback:** reply ya call outcome se time pehchan kar (jaise kal 5 baje) reminder khud set ho.
- Har reminder pehle **DNC list** aur quiet hours (9 PM se 9 AM) check kare, aur delivery status log kare.

7. Build order (original plan)

Step	Kaam	Result
1	Central DNC / consent list jo har channel bhejne se pehle check kare	Legal safety, ban ka risk kam
2	Delivery status webhooks (Twilio aur Meta)	Asli sent / delivered / read / failed status
3	Reply aate hi team ko instant alert (push + task)	Koi reply 24 ghante tak ignore nahi hoga
4	Inbound email reply detect + unsubscribe link + bounce handling	Email bhi follow-up engine ka hissa
5	Call outcome form aur callback reminder	Call ke baad agla step automatic

Step	Kaam	Result
6	Payment / booking link auto, sequence editor, reports	Conversion badhegi aur measure hogi
7	Reminder Engine (task, meeting, callback, invoice reminders)	Koi task, demo ya payment due miss nahi hoga

Note: Ye sab steps ab code mein complete hain. Live karne ke liye docs/COMMS_SETUP.md ka checklist follow karein.